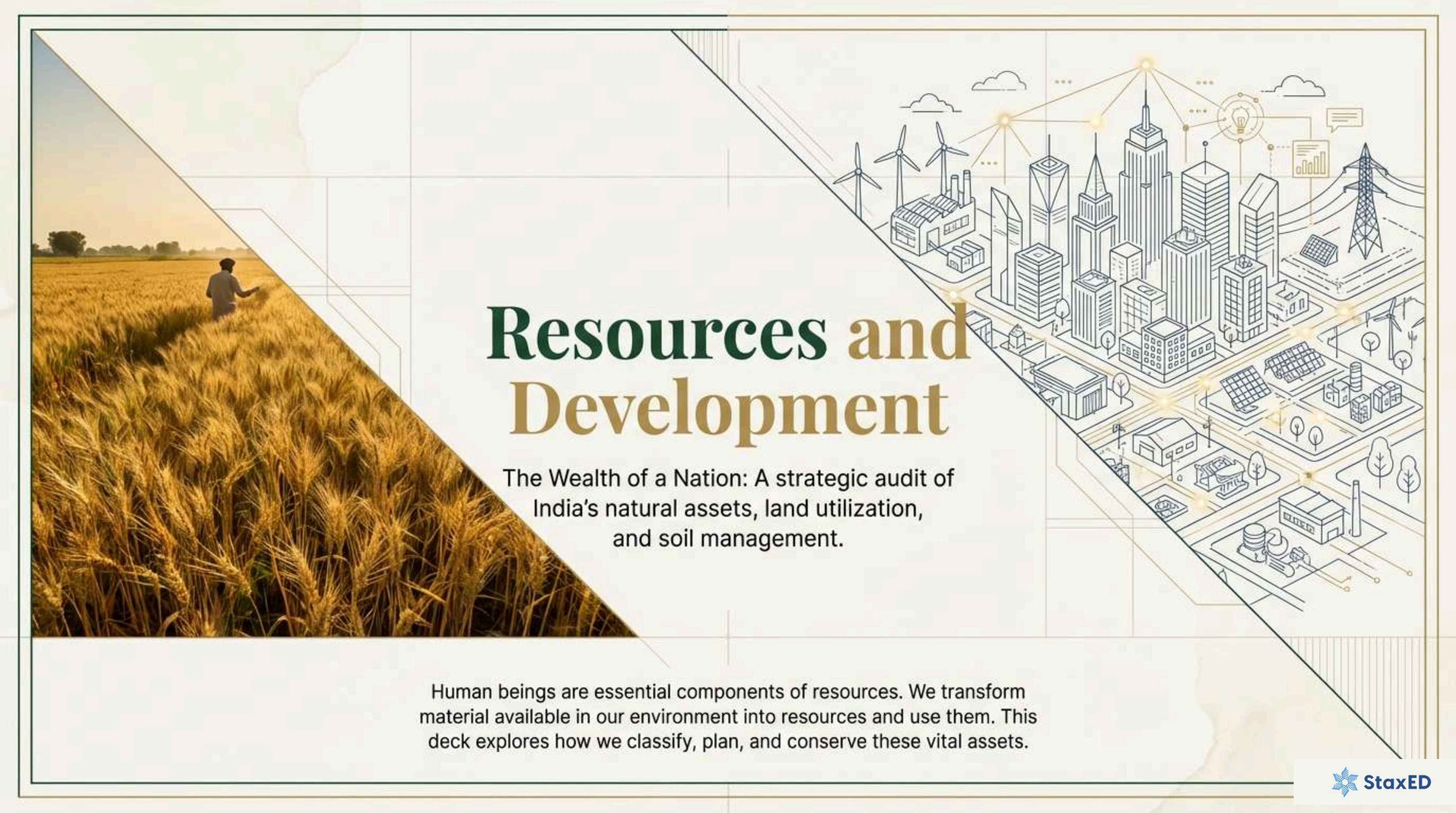




# Resources and Development

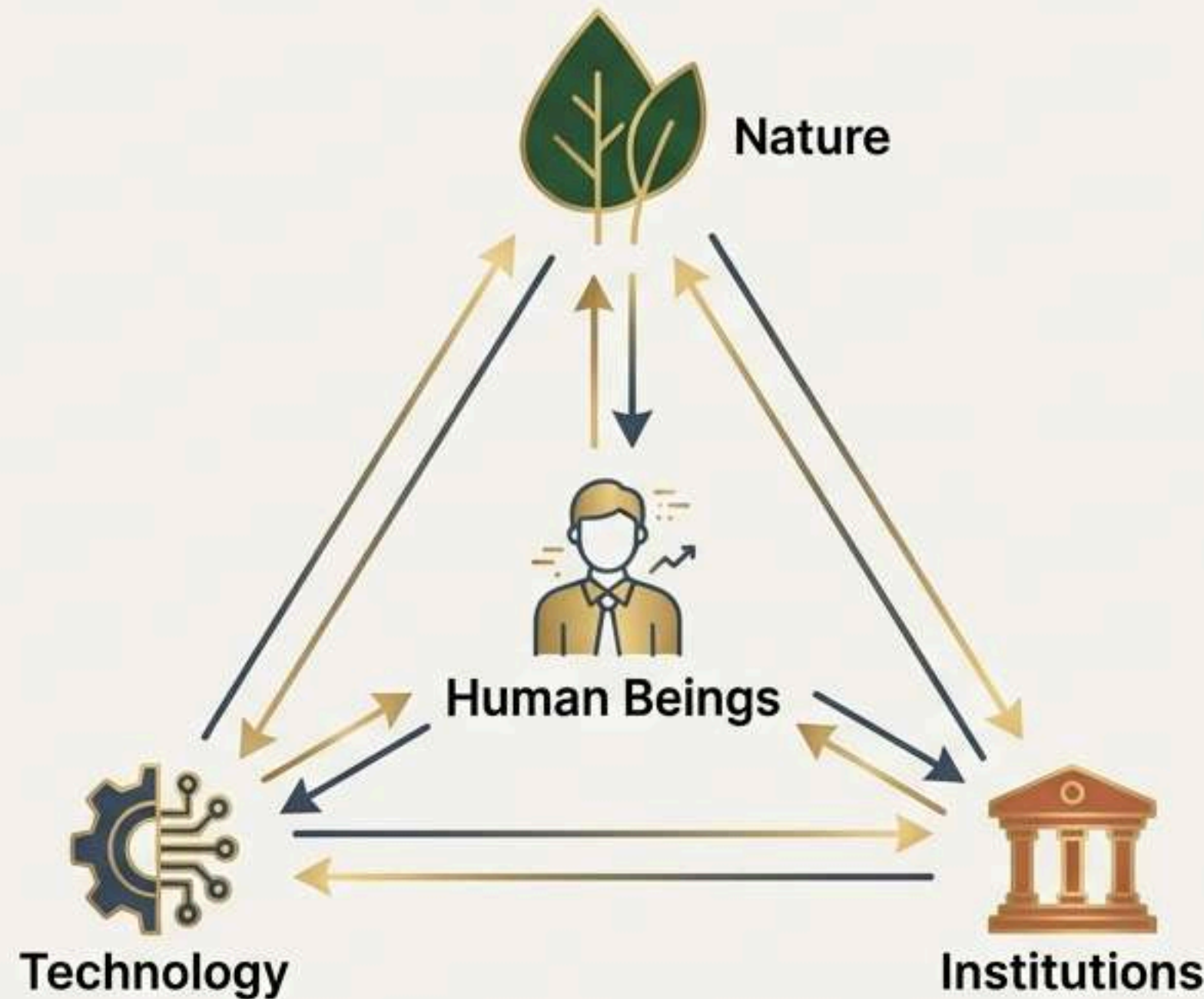
The Wealth of a Nation: A strategic audit of India's natural assets, land utilization, and soil management.

Human beings are essential components of resources. We transform material available in our environment into resources and use them. This deck explores how we classify, plan, and conserve these vital assets.



# Defining the Asset

The Interdependence of Nature, Technology, and Institutions



## The Criteria for a Resource

Everything in our environment which can be used to satisfy our needs is a "Resource" provided it is:

1. Technologically Accessible:  
We have the tools to reach it.
2. Economically Feasible: It  
pays to extract it.
3. Culturally Acceptable:  
Society approves its usage.



# The Classification Matrix





# Status of Development: An Inventory Audit



## Potential Resources

Found in a region but not utilized.

**Example Text:** Solar and wind energy potential in Rajasthan and Gujarat.



## Developed Resources

Surveyed; quality and quantity determined for utilisation.

**Note:** Depends on technology and feasibility.



## Stock

Potential to satisfy needs, but lack appropriate technology to access.

**Example Text:** Hydrogen as a specific energy source.



## Reserves

Subset of stock. We have the technical know-how, but usage hasn't started.

**Example Text:** Water in dams stored for future requirements.



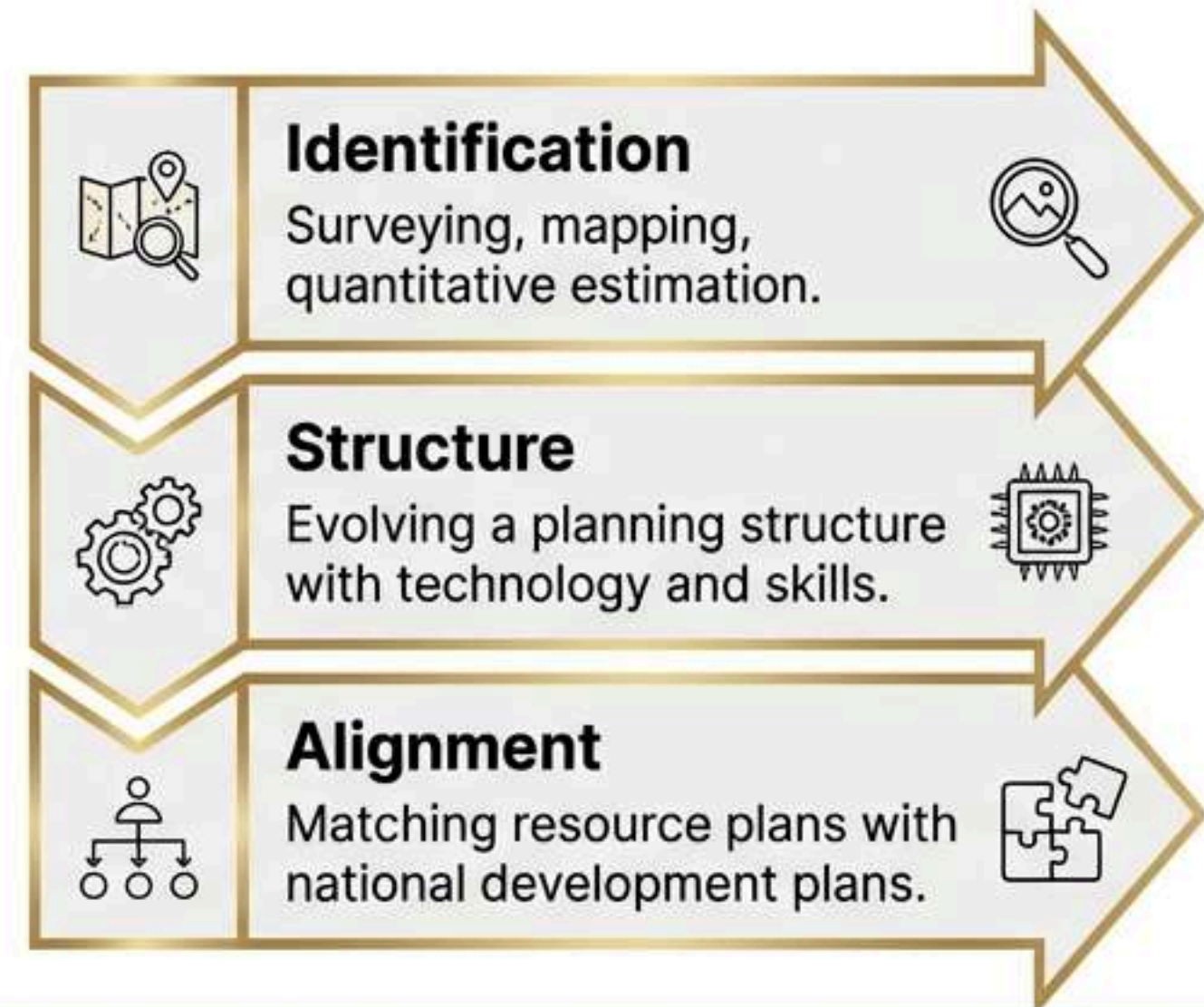
# The Crisis & The Strategy

## The Crisis: Indiscriminate Use

- **Depletion:** Satisfying the greed of a few.
- **Inequality:** Accumulation in few hands (Rich vs Poor).
- **Ecological Crises:** Global warming, ozone depletion, pollution.



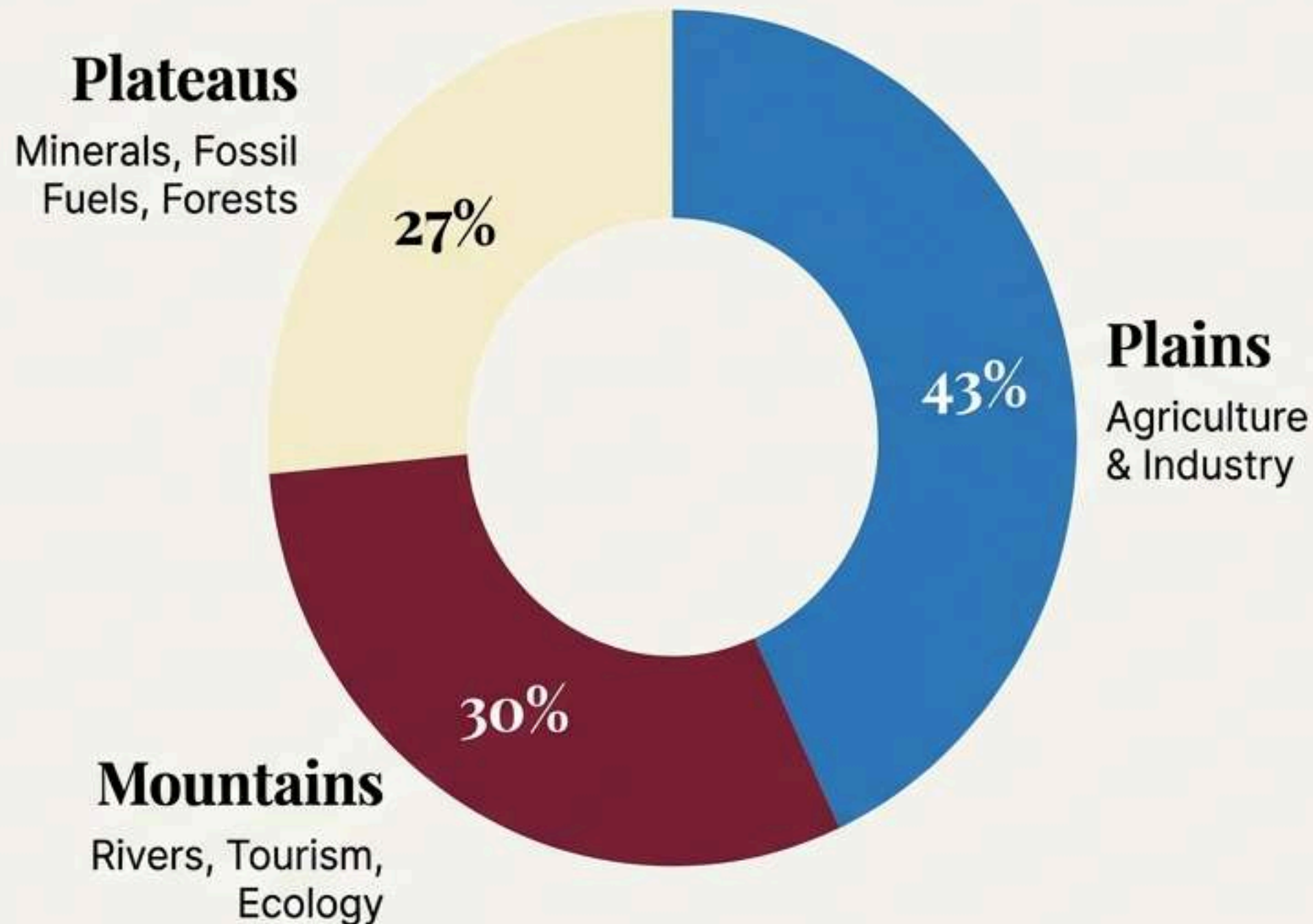
## The Solution: Resource Planning



**"Sustainable Economic Development:** Development should take place without damaging the environment, and present development should not compromise the needs of future generations."



# Land Resources: The Physical Portfolio



## Plains

Base for agriculture and industrial development.



## Mountains

Ensures perennial flow of rivers.

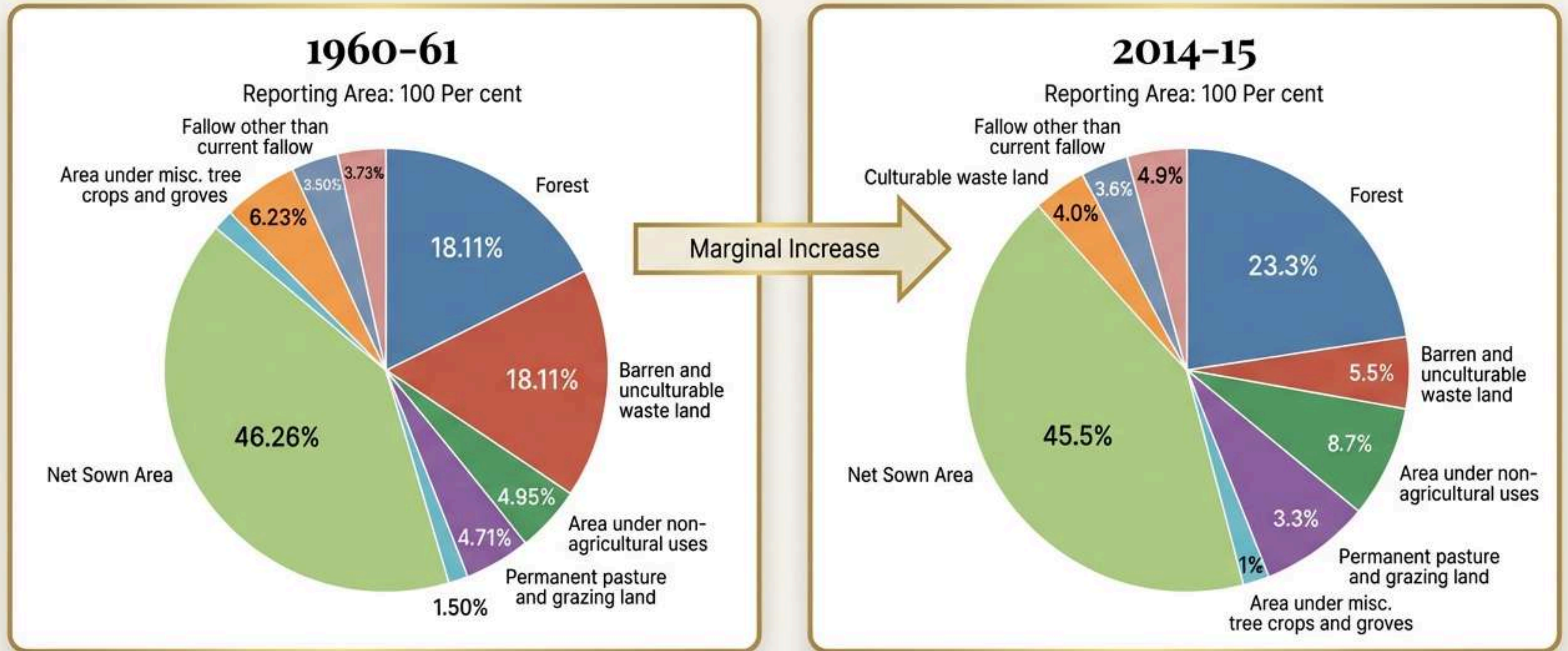


## Plateaus

The treasury of minerals and forests.



# Trends in Land Utilisation (1960–2014)



■ Forest      ■ Net Sown Area      ■ Permanent pasture and grazing land      ■ Fallow other than current fallow  
■ Barren and unculturable waste land      ■ Area under non-agricultural uses      ■ Area under misc. tree crops and groves      ■ Current fallow



# Land Degradation & Conservation

## The Threats (Degradation)



**Mining:** Deep scars and over-burdening (Jharkhand, Chhattisgarh).



**Overgrazing:** Major issue in Gujarat, Rajasthan, MP.



**Over-irrigation:** Salinity and logging (Punjab, Haryana).



**Industrial Effluents:** Water and land pollution.



## The Fixes (Conservation)



Afforestation and grazing management.



Planting of shelter belts.



Stabilisation of sand dunes (thorny bushes).



Treatment of industrial wastes.





# Soil: The Living Resource

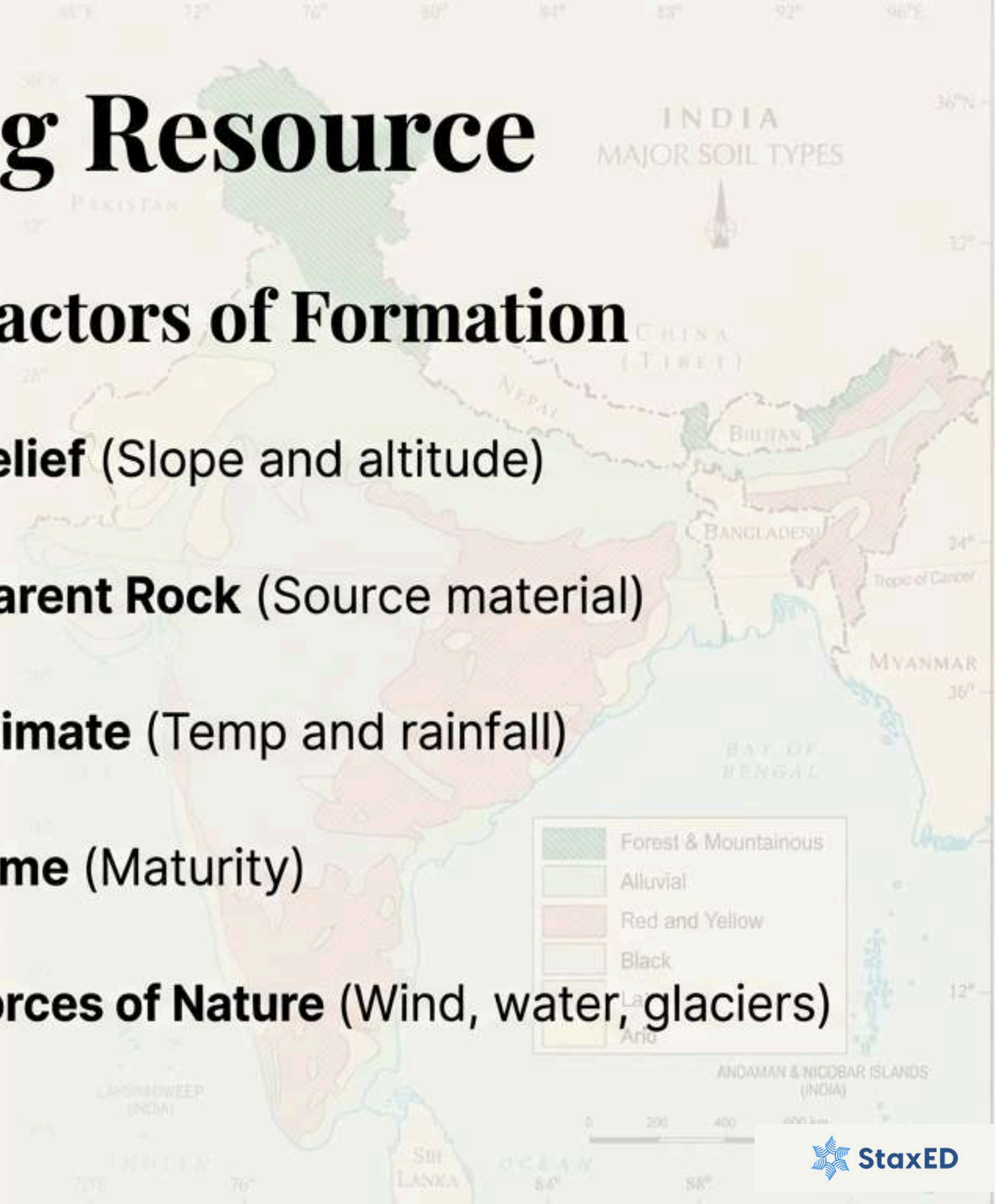
## Hero

### Definition

The medium of plant growth and support for living organisms. It takes millions of years to form soil up to a few centimetres in depth.

### Factors of Formation

1. **Relief** (Slope and altitude)
2. **Parent Rock** (Source material)
3. **Climate** (Temp and rainfall)
4. **Time** (Maturity)
5. **Forces of Nature** (Wind, water, glaciers)

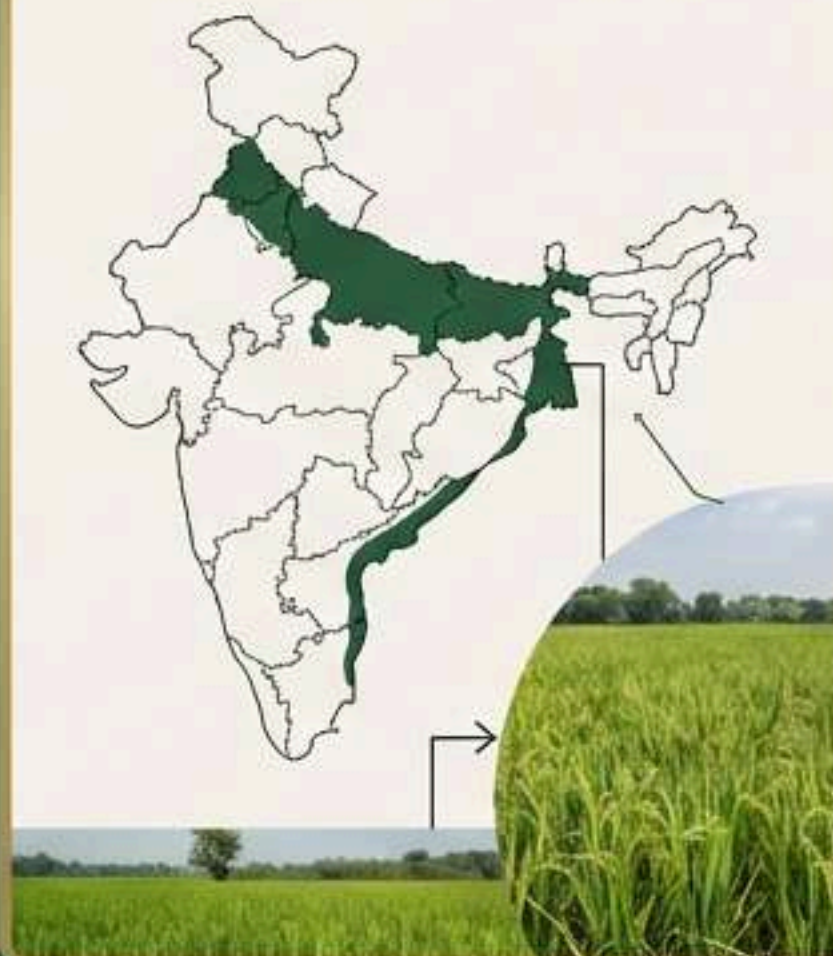




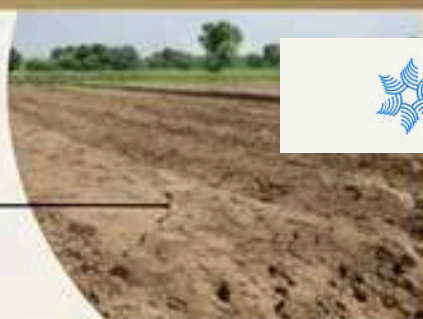
# The Agricultural Giants

Profile Card 1

## Alluvial Soil (The Builder)

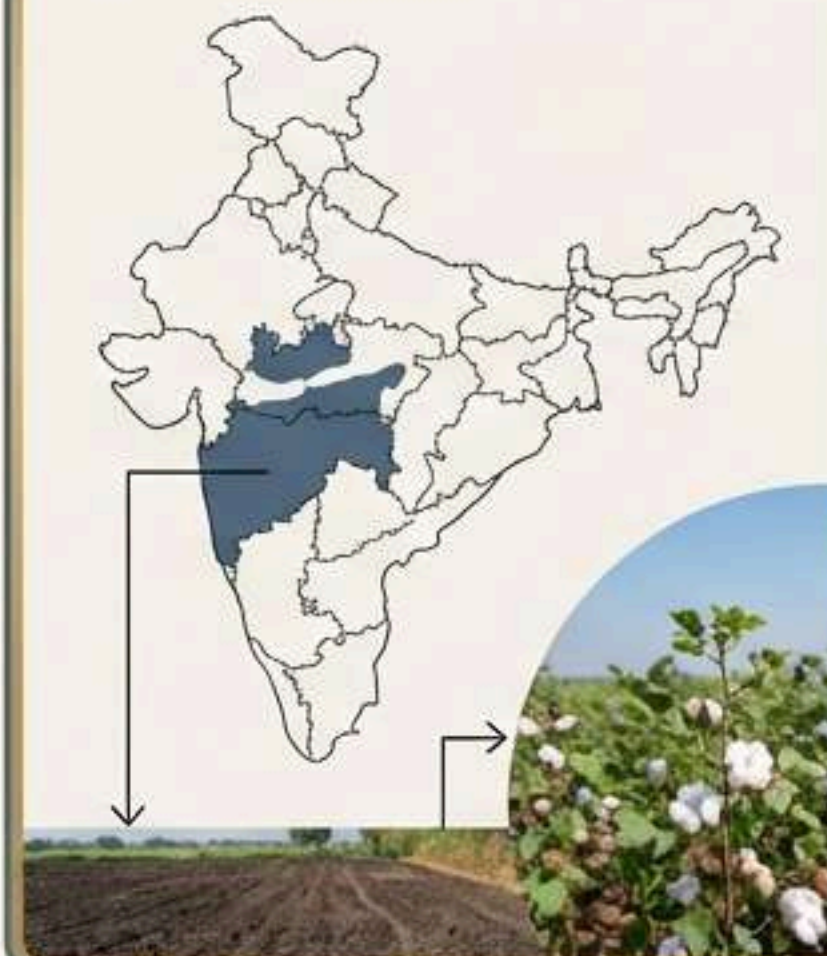


- **Composition:**  
Sand, silt, clay. Rich in potash, phosphoric acid, lime.
- **Types:**  
Khadar (New, fine, fertile) vs Bangar (Old, coarse, kankar nodules).
- **Crops:**  
Paddy, wheat, sugarcane, pulses.



Profile Card 2

## Black Soil / Regur (The Cotton Grower)



- **Features:**  
Made of lava flows. Clayey, moisture-holding, self-ploughing.
- **Nutrients:**  
Rich in calcium carbonate, magnesium, potash. Poor in phosphorus.
- **Crops:**  
Cotton.





# The Iron & The Brick

## Profile Card 1

### Red and Yellow Soils



Terracotta

Gold



- **Origin:**  
Crystalline igneous rocks in low rainfall areas.
- **Chemistry:**  
Reddish due to iron diffusion; Yellow when hydrated.
- **Location:**  
Odisha, Chhattisgarh, Piedmont zone of Western Ghats.

## Profile Card 2

### Laterite Soil (The Leached)



Slate Blue

Gold



- **Origin:**  
Intense leaching from heavy rain (Tropical).
- **Chemistry:**  
Acidic (pH < 6.0), low humus.
- **Utility:**  
Tea, Coffee, Cashew nuts (with manure).



# The Extremes

## Profile Card 1: Arid Soils



### Arid Soils

-  **Texture:** Sandy, saline, dry.
-  **Chemistry:** Lacks humus/moisture. High salt content.
-  **Barrier:** Lower horizons blocked by 'Kankar' (Calcium), restricting filtration.

## Profile Card 2: Forest Soils



### Forest Soils

-  **Location:** Hilly/Mountainous rainforests.
-  **Texture:** Loamy/silty in valleys; Coarse on upper slopes.
-  **Snow Areas:** Acidic with low humus due to denudation.

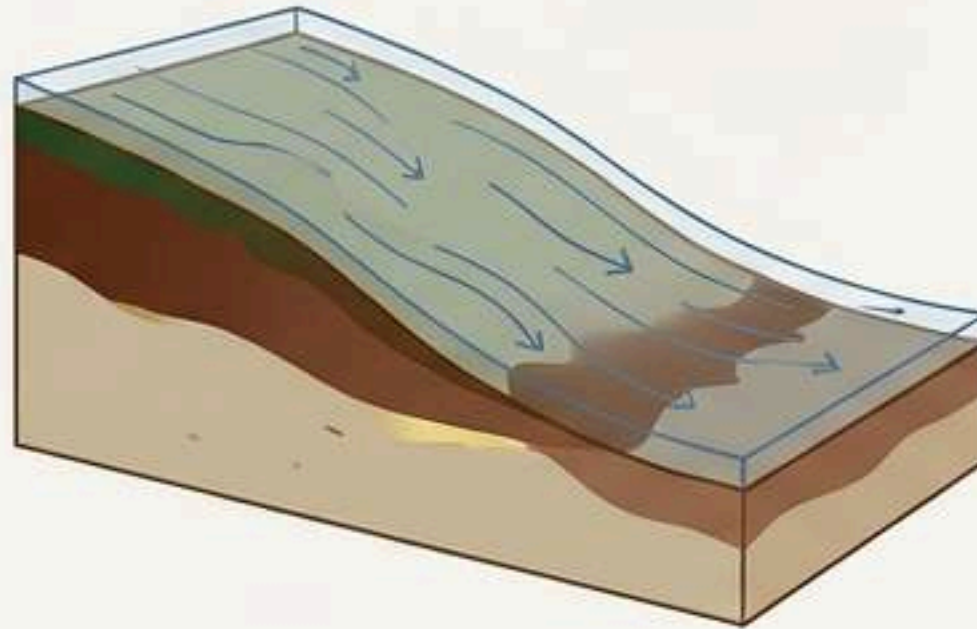


# Soil Erosion: The Mechanics of Loss



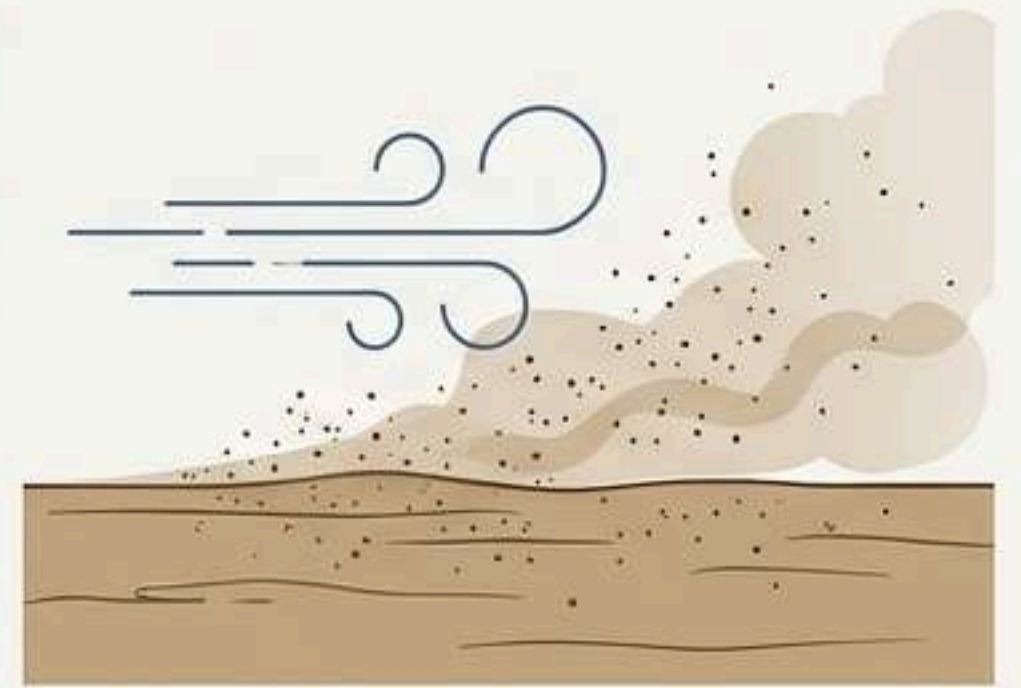
## Gully Erosion

Running water cuts deep channels (gullies) into clay. Creates "Bad land" unfit for cultivation.



## Sheet Erosion

Water flows as a sheet down a slope, washing away topsoil.



## Wind Erosion

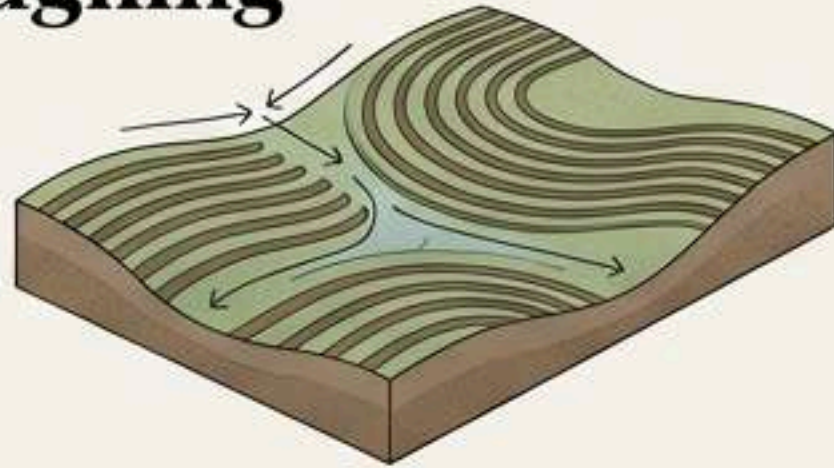
Loose soil blown off flat or sloping land.



# Soil Conservation: Engineering Nature

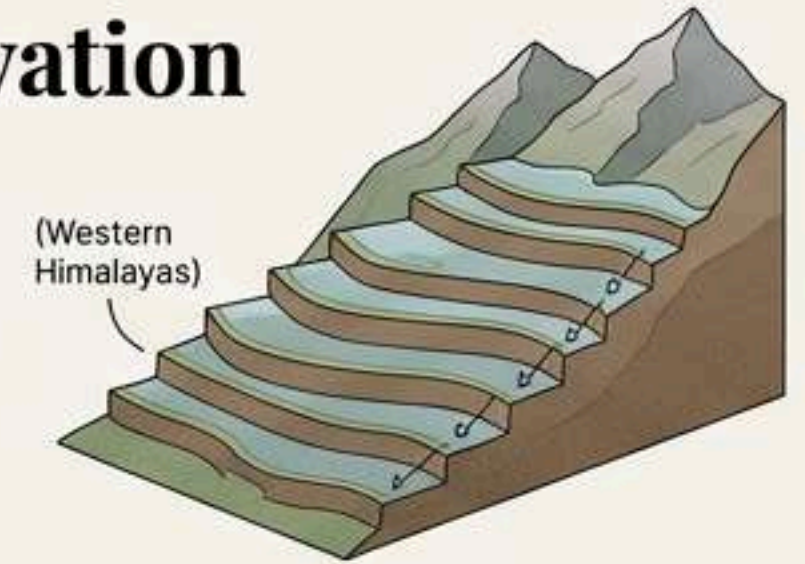
## Contour Ploughing

Decelerates water flow down slopes.



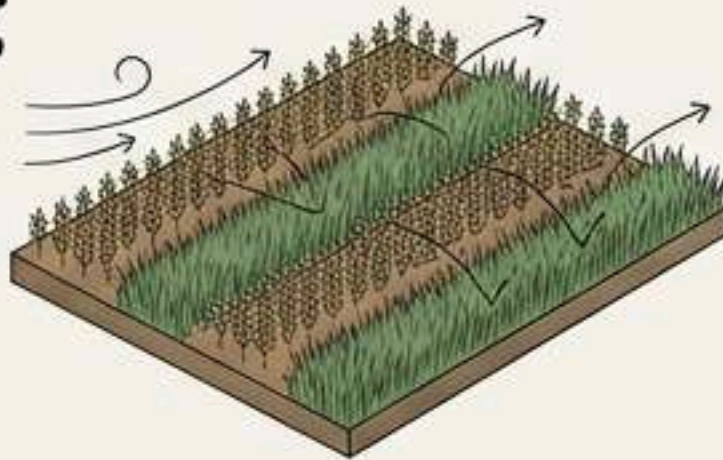
## Terrace Cultivation

Restricts erosion.



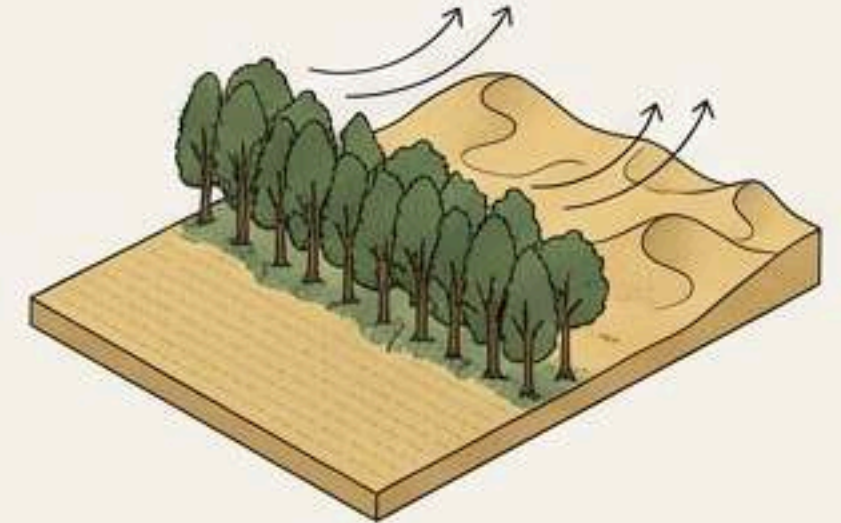
## Strip Cropping

Grass strips break the force of the wind.



## Shelter Belts

Stabilizes sand dunes and deserts.





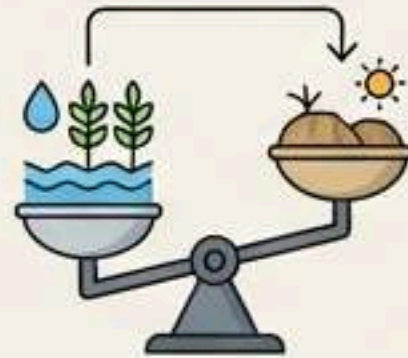
# Stewardship of the Nation's Wealth



Resources are not just nature; they are a function of technology, economy, and culture.



Planning is vital to balance regions of surplus with regions of shortage.



Land and soil are finite assets. Conservation is the only path to sustainable development.



Development cannot happen in isolation. The future of India's economic growth depends on the sustainable management of its physical inventory today.

Conservation is not just ecology; it is economy.